The interleaving distance between two interval modules

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I found nowhere the explicit computation of the interleaving distance between interval modules, tough it is well known that it equals the bottleneck distance.

1 Preliminaries

Let $\mathbf{k}$ be an arbitrary field, fixed throughout. We denote by $\mathbb{R}$ either the real line or the category induced by the poset $(\mathbb{R}, \leq)$. All functors are covariant.

1.1 The category of persistence modules

A **persistence module** is a functor $\mathbb{R} \to \mathrm{Vect}(\mathbf{k})$.

Example 1 (Important)**.** Let $I \subseteq \mathbb{R}$ be an interval. Define $\mathbf{k}^I \colon \mathbb{R} \to \mathrm{Vect}(\mathbf{k})$ by

$$t \mapsto \begin{cases} \mathbf{k} & \text{if } t \in I \\ \mathbf{0} & \text{else} \end{cases} \qquad \text{and} \qquad r \to s \ \mapsto \begin{cases} \mathbf{k} \xrightarrow{\mathrm{Id}} \mathbf{k} & \text{if } r, s \in I \\ \mathbf{0} & \text{else} \end{cases}$$

on objects and morphisms, respectively. Here $\mathbf{0}$ denotes the zero vector space of course. We call $\mathbf{k}^I$ the **interval module** over I.

Remark. Interval modules are the building blocks of persistence modules. In fact, one objective of persistence theory is to understand persistence modules by decomposing them into interval modules. However, this is not always possible. A situation when such a decomposition is guaranteed is when we work with finite dimentional vector spaces, that is, if we work over $\mathrm{vect}(\mathbf{k})$ instead of $\mathrm{Vect}(\mathbf{k})$.

The concise definition of persistence module can be expanded to aide comprehension. Let $V \colon \mathbb{R} \to \mathrm{Vect}(\mathbf{k})$ be a persistence module.

- For each $t \in \mathbb{R}$ and each arrow $r \to s$ denote $V(t) = V_t$ and $V(r \to s) = v_s^r$.
- Since V is a functor, we have the so-called **composition law**:

$$v_t^s \circ v_s^r = v_t^r \quad \text{whenever } r \leq s \leq t$$

and $V(t \to t) = v_t^t = \mathrm{Id}_{V_t}$.

Thus, a persistence module V is a family of vector spaces $(V_t)_{t\in\mathbb{R}}$ together with a family $(v_t^s\colon V_s \to V_t \mid s \leq t)_{s,t\in\mathbb{R}}$ of $\mathbf{k}$-linear maps that satisfy the composition law.

We got a set theorical definition from the categorical one above. In fact, the two are equivalent. We will use the first, but prefer the notation of the second.

Since persistence modules are functors, we can consider the natural transformations between them, and since those can be composed (using vertical composition) we obtain a category.

Definition 1. The **category of persistence modules** is $\mathbf{X} = \mathrm{Fun}(\mathbb{R}, \mathrm{Vect}(\mathbf{k}))$.

Yes, this category is fun. Concretely, a morphism of persistence modules $\alpha\colon U \to V$ is a family of linear maps $(\alpha_r\colon U_r \to V_r)_{r\in\mathbb{R}}$ such that the diagram

$$\begin{tikzcd} U_s \arrow[r, "u_t^s"] \arrow[d, "\alpha_s"'] & U_t \arrow[d, "\alpha_t"] \\ V_s \arrow[r, "v_t^s"] & V_t \end{tikzcd} \tag{1}$$

commutes for all $s \leq t$, i.e., $v_t^s \circ \alpha_s = \alpha_t \circ u_t^s$.

Explain composition with a diagram.

1.2 The shift functor

Let U be a persistence module and fix any $t \in \mathbb{R}$. Define a new persistence module $U[t]$ by

$$U[t](r) = U_{t+r} \quad \text{and} \quad U[t](r \to s) = u_{t+r}^{t+s}.$$

We say that $U[t]$ is a **shifted module**. Note that

$$U[0] = U \quad \text{and} \quad U[t][t'] = U[t+t']$$

for any $t, t' \in \mathbb{R}$. Similarly, for any morphism of persistence modules $\alpha\colon U \to V$, we can define a **shifted module morphism** $\alpha[t] = (\alpha_{r+t})_{r\in\mathbb{R}}$.

Suppose we've made two shifts of U, say $U[s]$ and $U[t]$, where $s \leq t$. We might ask ourserlves: *what is the relationship between these shifted modules?* In other words, we wonder if there is a canonical natural transformation $\alpha\colon U[s] \to U[t]$. To find out, consider any $p \to q$ and apply both $U[s]$ and $U[t]$ to this arrow. We get

$$U[s](p \to q) \;=\; U_{p+s} \xrightarrow{u_{q+s}^{p+s}} U_{q+s}$$

and

$$U[t](p \to q) \;=\; U_{p+t} \xrightarrow{u_{q+t}^{p+t}} U_{q+t}.$$

Since $\alpha = (\alpha_r)_{r\in\mathbb{R}}$ is determined by its components, we need to find the vertical arrows that make the square

$$\begin{tikzcd}
U_{p+s} \arrow[r, "u_{q+s}^{p+s}"] \arrow[d, "\alpha_p"'] & U_{q+s} \arrow[d, "\alpha_q"] \\
U_{p+t} \arrow[r, "u_{q+t}^{p+t}"'] & U_{q+t}
\end{tikzcd}$$

commute. Set $\alpha_p = u_{p+t}^{p+s}$ and $\alpha_q = u_{q+t}^{q+s}$. Then the composition law give us

$$u_{q+t}^{p+t} \circ u_{p+t}^{p+s} = u_{q+t}^{p+s} = u_{q+t}^{q+s} \circ u_{q+s}^{p+s},$$

which means the square commutes. So, yes there is a canonical way to map $U[s]$ to $U[t]$. Denote $\alpha = U[s \to t]$ and define

$$U[s] \xrightarrow{U[s\to t]} U[t] \quad = \quad \left(u_{r+t}^{r+s}\right)_{r\in\mathbb{R}}.$$

Remark 1.

- For $t \geq 0$, we denote $\mathbb{1}_U^t = U[0 \to t]$.
- Observe that

  $$\mathbb{1}_U^0 = U[0 \to 0] = (u_r^r)_{r\in\mathbb{R}} = (\mathrm{Id}_{U_r})_{r\in\mathbb{R}} = \mathbb{1}_U$$

  is an identity natural transformation. So, for $t \geq 0$, we might think of $\mathbb{1}_U^t$ as a generalized identity.
- By definition, we get $U[t \to t] = \mathbb{1}_{U[t]}$ for any $t \in \mathbb{R}$. Moreover, if $r \leq s \leq t$, for any $p \in \mathbb{R}$ we have

  $$\begin{aligned}
  (U[s \to t] \circ U[r \to s])_p &= U[s \to t]_p \circ U[r \to s]_p \\
  &= u_{p+t}^{p+s} \circ u_{p+s}^{p+r} \\
  &= u_{p+t}^{p+r} \\
  &= U[r \to t]_p
  \end{aligned}$$

  thanks to the composition law. Thus

  $$U[s \to t] \circ U[r \to s] = U[r \to t].$$

The last remark leads us to the following definition.

Definition 2. The **shift functor** $T = [\,\cdot\,] \colon \mathbb{R} \to \mathrm{End}(X)$ is defined as follows

(i) Objects: for each $t \in \mathbb{R}$ we have a functor $T(t)\colon X \to X$ denoted T_t, which is defined by $T_t(U) = U[t]$ on objects $U \in X$, and by $T_t(\alpha) = \alpha[t] = (\alpha_{r+t})_{r\in\mathbb{R}}$ on morphisms $\alpha = (\alpha_r)_{r\in\mathbb{R}}$. (Equivalently, we can define $T_t(\alpha) = \alpha \star [t]$, using a whiskering)

(ii) Morphisms: for each $r \to s$, define a natural transformation $T(r \to s)\colon T_r \to T_s$ by

$$T(r \to s) = (U[r \to s])_{U\in X}\,.$$

1.3 The interleaving distance between persistence modules

Let U and V be persistence modules. A **t-interleaving** of U and V is a pair of natural transformations ($\phi\colon U \to V[t]$, $\psi\colon V \to U[t]$) that make the diagrams

$$\begin{tikzcd} U \arrow[rr, "\mathbb{1}_U^{2t}"] \arrow[dr, "\phi"'] & & U[2t] \\ & V[t] \arrow[ur, "\psi[t]"'] & \end{tikzcd} \qquad \begin{tikzcd} V \arrow[rr, "\mathbb{1}_V^{2t}"] \arrow[dr, "\psi"'] & & V[2t] \\ & U[t] \arrow[ur, "\phi[t]"'] & \end{tikzcd}$$

commute. If such a pair exists, we say that U and V are t-interleaved.

Example 2. Two persistence modules are isomorphic if and only if they are 0-interleaved. Thus, we can think of interleavings as generalized isomorphisms. Let us see why...

Definition 3. The **interleaving distance** between two persistence modules U and V is

$$d_T(U, V) = \inf \{t \geq 0 \mid \text{there exists a } t\text{-interleaving of } U \text{ and } V\}\,.$$

in case there is at least one interleaving of U and V. If no such interleaving exists, we set $d_T(U, V) = \infty$.

The subscript T remind us that d_T depends on the shift functor. On the other hand, d_T is not an actual metric, but a *pseudometric*, since $d_T(U, V)$ does not imply $U \cong V$.

Example 3. An example that proves the last sentence.

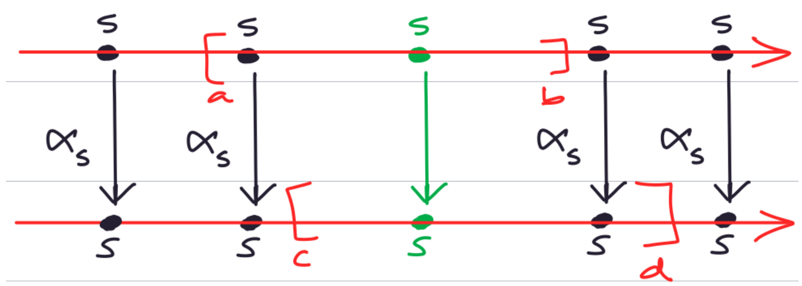
Let us prove d_T satisfies the triangle inequality.

2 The computation

We'll do the computation for the case of closed intervals. Let $I = [a, b]$ and $J = [c, d]$. Denote by $\mathcal{I} = \mathbf{k}^I$ and $\mathcal{J} = \mathbf{k}^J$ the associated interval modules.

Our strategy is to find some apropriate bounds for $d_T(U, V)$. Befor going into the search for the pair of morphisms that would make the triangles above commute, we need to know how are the morphisms between arbitrary interval modules, because then we can characterize the morphisms between shifted interval modules.

Let $\alpha\colon \mathcal{I} \to \mathcal{J}$ be any persistence module morphism. For each $s \in \mathbb{R}$ we have a map of vector spaces $\alpha_s\colon \mathcal{I}_s \to \mathcal{J}_s$. The only available options for $\mathcal{I}_s$ and $\mathcal{J}_s$ are either $\mathbf{k}$ or $\mathbf{0}$.



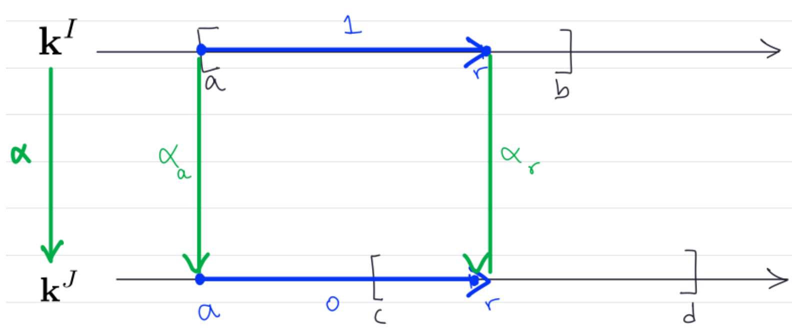
Note that, if $s \notin I \cap J$, then either $\mathcal{I}_s = \mathbf{0}$ or $\mathcal{J}_s = \mathbf{0}$, but in any case α_s is the zero map. Thus, we always get trivial morphisms outside of the intersection. Inside the intersection, when $s \in I \cap J$, we have a linear map $\alpha_s\colon \mathbf{k} \to \mathbf{k}$, but since $\mathbf{k}$ is a field, α is nothing more than multiplication by some scalar $\lambda \in \mathbf{k}$, so $\alpha_s\colon x \mapsto \lambda x$. Such λ could be zero, however.

As a first result, we have the following.

Fact 1. If $I \cap J = \varnothing$, then $\operatorname{Hom}(\mathbf{k}^I, \mathbf{k}^J) = \mathbf{0}$.

With this trivial case out of the way, assume $I \cap J \neq \varnothing$, and let us consider two cases: $a < c$ or $c \leq a$.

Case 1. Suppose $a < c$. Recall that our goal is to characterize the morphisms $\mathbf{k}^I \to \mathbf{k}^J$. Since we already know what happens when $t \notin I \cap J$, let us focus on $\alpha_t\colon \mathbf{k}^I_t \to \mathbf{k}^J_t$ with $t \in I \cap J$. Take the arrow $a \to r$ and apply



both $\mathbf{k}^I$ and $\mathbf{k}^J$. We get We know $\mathbf{k}^I(a \to r) =$ By (1) The conclusion of this case is that $\operatorname{Hom}(\mathbf{k}^I, \mathbf{k}^J) \cong \mathbf{0}$, even if $I \cap J \neq \varnothing$.

Case 2. Suppose $c \leq a$, meaning that J *starts* before I. Now we consider two more cases.

Case 2.1. Suppose $b < d$. We also get $\mathrm{Hom}(\mathbf{k}^I, \mathbf{k}^J) \cong \mathbf{0}$. the problem was that J *died* after I.

Case 2.2. Suppose $d \leq b$. Here we prove there is an isomorphism $\mathrm{Hom}(\mathbf{k}^I, \mathbf{k}^J) \cong \mathbf{k}$.

Now we come to the following conclusion.

Fact 2. If $c \leq a \leq d \leq b$, then $\mathrm{Hom}(\mathbf{k}^I, \mathbf{k}^J) \cong \mathbf{k}$.

Both Fact 1 and Fact 2 will help us determine $d_T(\mathbf{k}^I, \mathbf{k}^J)$ for any closed intervals I and J.

2.1 Bound above

We start looking for values of $t \geq 0$ for which a t-interleaving of $\mathcal{I} = \mathbf{k}^I$ and $\mathcal{I} = \mathbf{k}^J$ exists, that is, for which we can find maps $\phi\colon \mathcal{I} \to \mathcal{I}[t]$ and $\psi\colon \mathcal{I} \to \mathcal{I}[t]$ that make the diagrams

$$
\begin{tikzcd}
\mathcal{I} \arrow[rr, "\mathbb{1}_{\mathcal{I}}^{2t}"] \arrow[dr, "\phi"'] & & \mathcal{I}[2t] \\
& \mathcal{I}[t] \arrow[ur, "\psi{[t]}"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{I} \arrow[rr, "\mathbb{1}_{\mathcal{I}}^{2t}"] \arrow[dr, "\psi"'] & & \mathcal{I}[2t] \\
& \mathcal{I}[t] \arrow[ur, "\phi{[t]}"'] &
\end{tikzcd}
$$

commute.

2.2 Bound below